

Diagnosing Conformal Prediction Failures Under Distribution Shift

A COVID-19 Case Study

Chorok Lee · KAIST **UAI 2026 — Amsterdam, Netherlands**

Top-1 SHAP Concentration Flags GBM CP Risk Pre-deployment signal for multiclass APS conformal prediction.

$\rho = 0.853$
16-task correlation

40%
exploratory cutoff

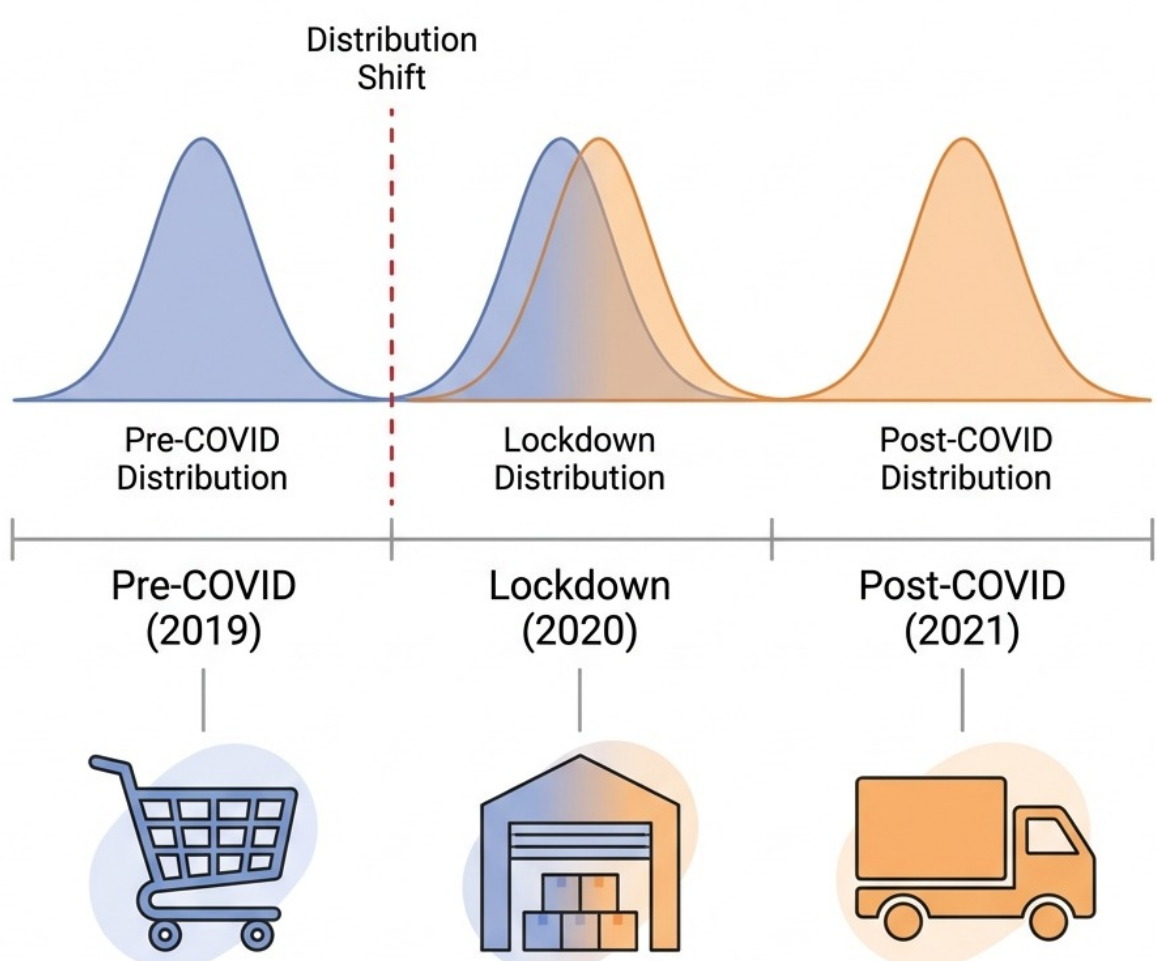
91.7%
12-check audit

Problem & Motivation

Conformal Prediction (CP): distribution-free uncertainty; true label in prediction set with $\text{prob} \geq 1 - \alpha$ (e.g., 90%).
Problem: shift can break coverage.

Research Question:

Can we **predict** when CP fails **before** deployment?



Key Insight

Models with **concentrated feature reliance** are at risk:

- Single feature drift $\Rightarrow$ large impact
- Less redundancy across features
- APS scores inflate after shift

High concentration
 $\downarrow$
Risk flag

SALT Case Study & Analysis Set

RelBench Supply Chain Benchmark

- **8 classification tasks**
- **COVID-19 temporal split**
- Feb–Jul 2020 (onset $\rightarrow$ peak)
- LightGBM multiclass APS conformal prediction

Primary correlation adds **8 external multiclass tasks**, yielding **16 tasks across 9 domains**.

Why Not Standard Shift Tests?

MMD, C2ST, and PSI detect dataset shift. They detect shift, not CP failure severity.

$\rho \leq 0.19$
weak relation to SALT coverage drop

Interpretation: SHAP concentration asks whether one fragile model direction already dominates.

Keywords

- CP** Conformal Prediction; builds prediction sets with validity guarantees.
- Coverage** Fraction of samples where true label is in the prediction set (target: 90%).
- APS** Adaptive Prediction Sets; ranks labels by model probability. The score says how far down the list the true label appears.
- SHAP** SHapley Additive exPlanations; feature attribution for concentration.
- SALT** Supply chain ALlocAtion; RelBench supply-chain benchmark.
- GBM** Gradient Boosting Machine; LightGBM/XGBoost/CatBoost family.
- MLP** Multilayer Perceptron; neural-network baseline.

MAIN RESULT

Pre-deployment risk signal predicts coverage drop LightGBM/GBM multiclass APS conformal predictors under shift

$\rho = 0.853$
Spearman Correlation
 $p < 0.001$ | $n = 16$ tasks | 9 domains

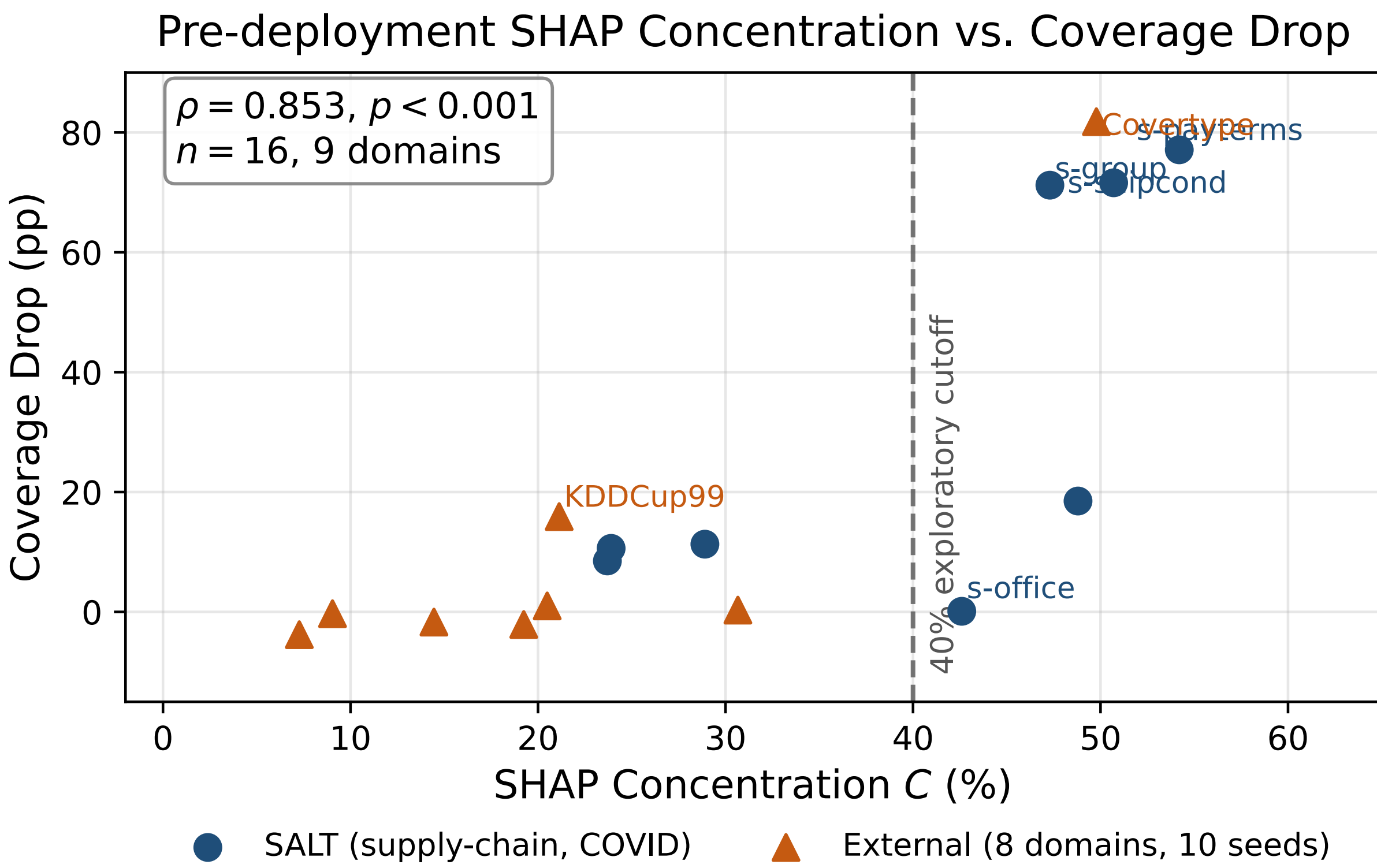


Fig. 1: Each point is a task. C is measured before deployment; coverage drop = validation coverage – post-test test coverage, in percentage points. The dashed line is the exploratory SALT-derived cutoff.

Takeaway: higher concentration tracks larger CP coverage loss.
Across 16 tasks, high- C models occupy the high-drop region.

Mechanism & Theory

Top-1 concentration: share of mean absolute SHAP importance

$$C = \frac{\max_j \mathbb{E}|\phi_j|}{\sum_{j=1}^p \mathbb{E}|\phi_j|}$$

C = concentration, ϕ_j = SHAP value for feature j , p = number of features

Idealized theorem: concentration amplifies APS score inflation

$$\mathbb{E}[s_{\text{test}}] \geq C(1 - (K-1)\epsilon) + (1-C)(1 - (K-1)\bar{h})$$

APS score = how far down the ranked label list we go to reach the true label. If the dominant shifted feature weakens true-class support ($\epsilon < \bar{h}$), high C raises that score; with the pre-shift CP cutoff fixed, coverage drops.

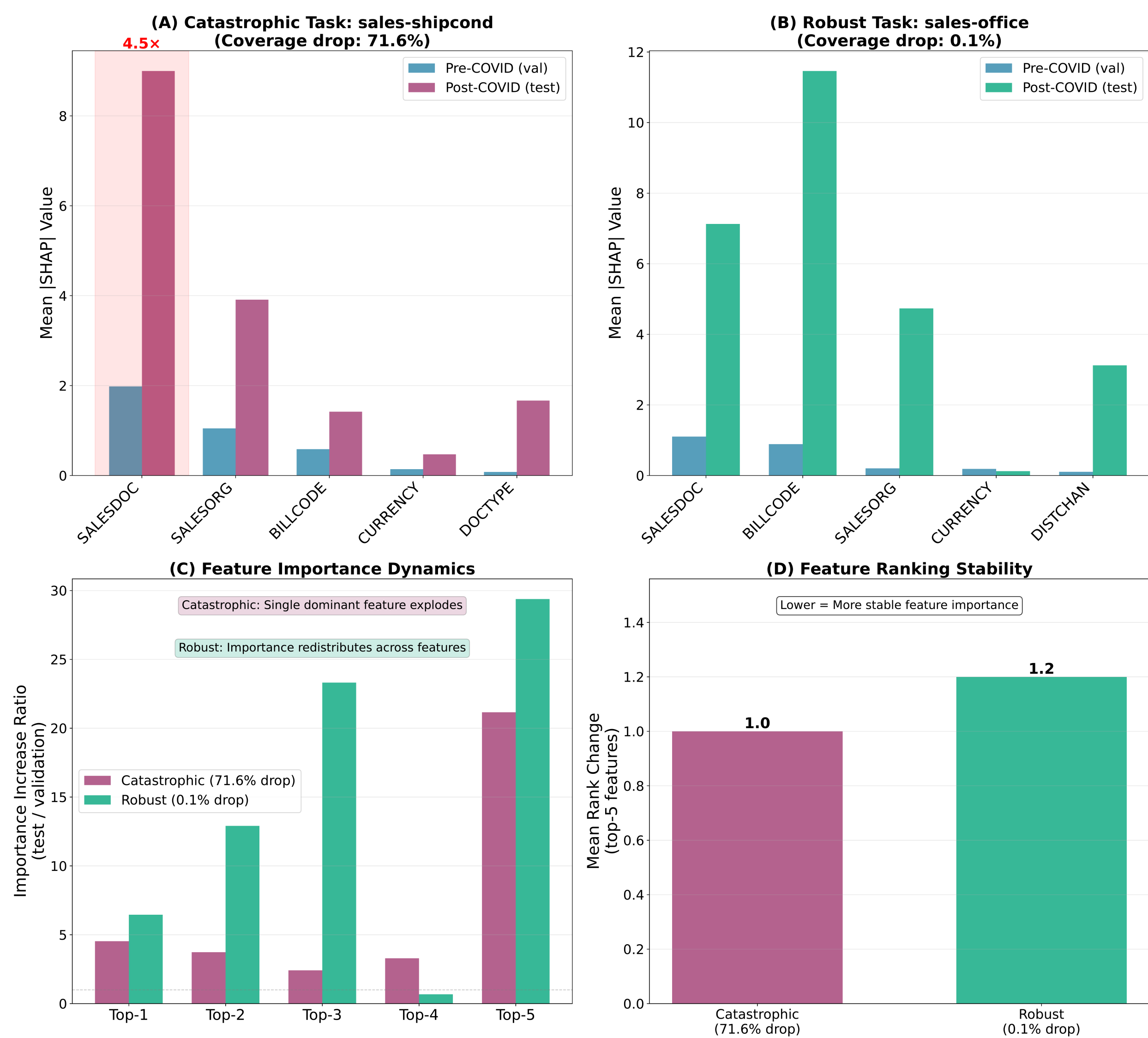


Fig. 2: Catastrophic failure keeps SHAP mass concentrated on one shifted feature; the robust example redistributes importance across several top features.

- A. Catastrophic (77pp drop):** SHAP mass jumps to one feature.
- B. Robust (8pp drop):** importance shared across top features.
- C. Ratio:** failing task has a top-feature spike.
- D. Stability:** catastrophic locks onto one feature (stable but fragile); robust redistributes (less stable but resilient).

Mechanism: high- C GBMs can create a single point of failure.

The claim is directional: stable/protective top features can be exceptions.

Specificity Check: Low- C Holds Out

Low- C checks avoid observed catastrophic loss.

Cutoff predicts low risk; observed drop validates it.

Dataset	Top-1 C	Obs. drop	Rule
CivilComments	26.2%	-0.7pp	No flag
Amazon	1.9%	+0.4pp	No flag
Covertypes*	12.5%	+8.0pp	No flag
Adult	27.5%	+1.4pp	No flag

No flag = $C < 40\%$; max observed drop is +8.0pp.

* Different split from Fig. 1 Covertypes ($C = 49.8\%$, 81.8pp drop).

91.7%

11/12 rule labels matched
8 SALT + 4 specificity checks

Scope & Model Sensitivity

Model	ρ	Why
GBM	0.55–0.83	Headline $\rho = 0.853$ is 16-task LightGBM.
RF	0.30	Bagging dilutes single-feature dependence.
MLP	0.43	Failures may be distributed/global; needs neural diagnostics.

Scope: GBM multiclass APS CP. SHAP concentration captures concentrated dependence failures, not every failure mode.

Conclusions & Future Work

Takeaways:

- Top-1 SHAP concentration predicts APS CP coverage loss in GBMs: $\rho = 0.853$ ($p < 0.001$)
- The 40% cutoff is exploratory: **91.7%** label agreement across 12 checks (11/12)

Future Directions:

- **Prospective validation:** test the cutoff on new domains
- **Top-N concentration:** explore top-2, top-3 variants
- **Protective factors:** detect stable top features that prevent failure
- **Neural diagnostics:** gradient-based explanations for MLPs

Open Questions

- Can we design models that are **intrinsically low-concentration**?
- Does the **40% threshold** hold prospectively across new domains?
- When does high concentration stay safe because the top feature is stable?
- Can **streaming CP** monitor concentration and trigger mitigation before failure?

Paper, Code & Contact

Repository

github.com/ChorokLeeDev/
conformal-covid-uai2026

Contact

choroklee@kaist.ac.kr



Scan for paper & code